

PhD Candidacy Oral Talk

Verification, Behavior, and Scale: An Agentic Framework for Discrete Event Traffic Simulation

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Date: Monday, July 13, 2026 : 4:00 p.m

THE WRITTEN QUESTIONS AND FOLLOWUPS

Q1. AGENTIC EXPERIMENTATION AND V&V

How can AutoTrafficSim autonomously generate and refine DES experiments while preserving scientific validity?

Q1. Follow-up: Stopping logic, quarantine, audit and pruning, and the human-agent boundary.

Q2. BEHAVIORAL ASSUMPTIONS AND SCIENTIFIC CREDIBILITY

How are unobserved evacuation behaviors modeled and justified, and what does AutoTrafficSim add beyond SUMO?

Q2. Follow-up: Parameter ranges, limited-data credibility, and a standard SUMO benchmark.

Q3. HYBRID MESO-MICRO TRAFFIC SIMULATION

How can mesoscopic and microscopic models be integrated and synchronized for effective large-scale simulation?

Q3. Follow-up: Interface diagrams, North Los Angeles, and whether PDES is fundamental or primarily needed for speed.

The Rise of Scientific AI Agents

Towards Scientific Intelligence: A Survey of LLM-based Scientific Agents

Shuo Ren^{*1,2}, Can Xie^{*1,3}, Pu Jian^{1,3}, Zhenjiang Ren^{1,3},
Chunlin Leng^{1,3}, Jiajun Zhang^{†1,2,3,4}

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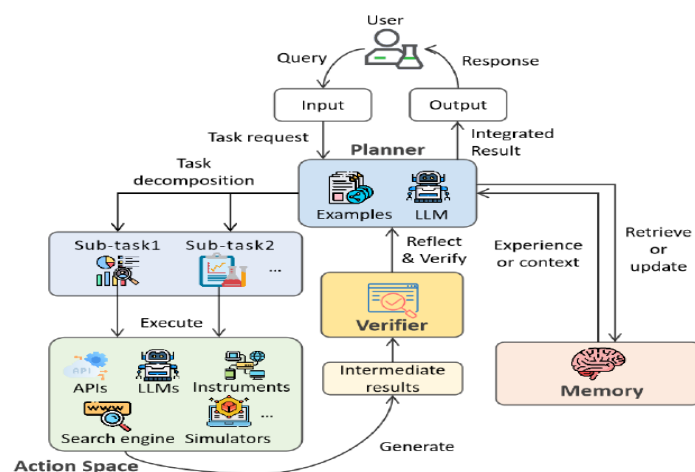


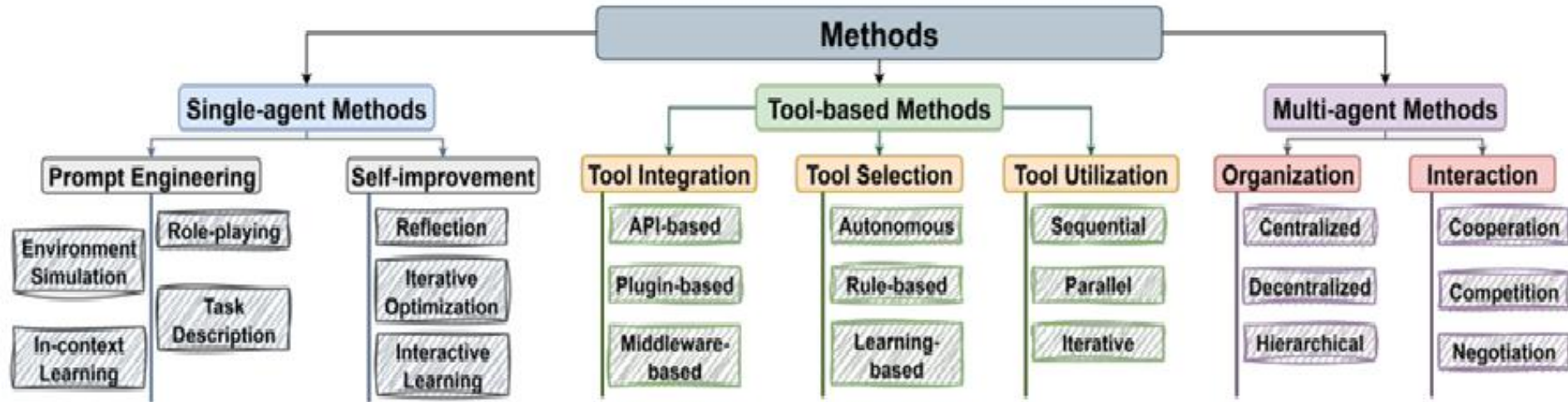
Figure 1: A typical architecture of LLM-based scientific agents. Note that in mainstream agent frameworks, planners are predominantly implemented based on LLMs, and their capabilities include task planning, reflection, and verification, etc. For the sake of abstraction, we represent these functions with a single planner in this architecture diagram. However, in specific implementations, different agents might be set up to accomplish distinct functions (see Section 2.1 for further discussion about planner types).

“LLMs are now evolving into LLM-based scientific agents that automate critical tasks, ranging from hypothesis generation and experiment design to data analysis and simulation.” Ren et al., 2026.

Some Agentic Simulation 2026 Papers

- Electrolyte discovery by agentic MD simulation (AI4Mat-ICLR 2026)
- Aitomia: An Agentic Framework for AI-Driven Atomistic and Quantum Chemical Simulations Journal of Chemical Theory and Computation (ACS) Peer-reviewed, published June 2026
- QUASAR: A Universal Autonomous System for Atomistic Simulation and a Benchmark of Its Capabilities Journal of Chemical Information and Modeling (ACS) Peer-reviewed, published May 2026
- GENIUS: An Agentic AI Framework for Autonomous Design and Execution of Simulation Protocols Communications Materials (Nature Portfolio) published April 2026

Agentic Taxonomy and Methods



Single-agent methods

- One LLM Agent through prompting and self-improvement.

Tool-based methods

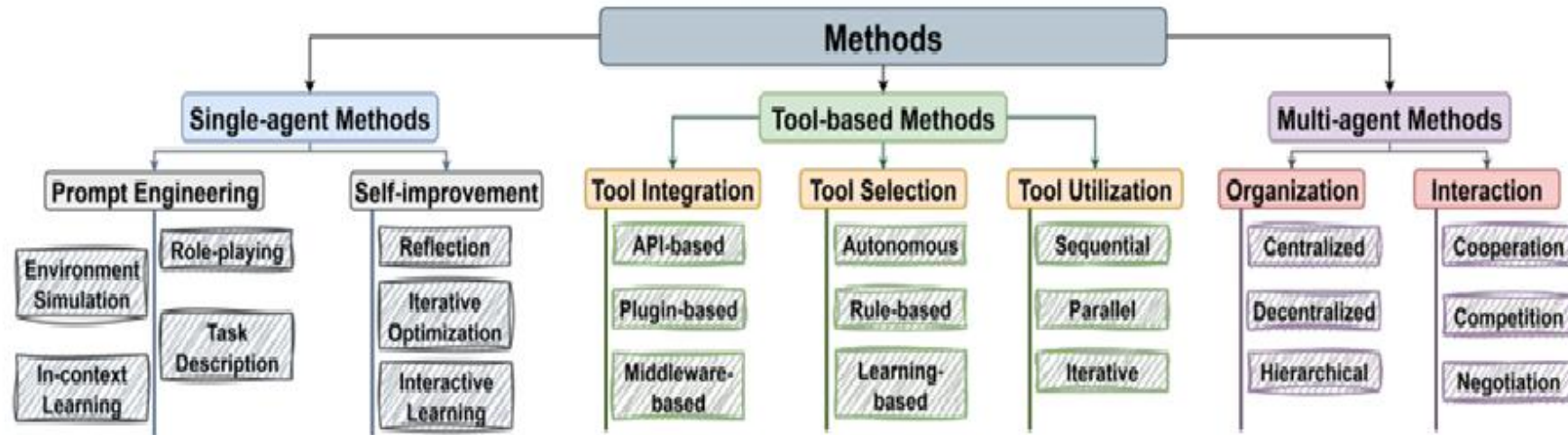
- Extend the agent's reasoning through external tools.

Multi-agent methods

- Coordinate multiple agents through organization and interaction.

Zhao et al., LLM-based Agentic Reasoning Frameworks: A Survey from Methods to Scenarios, 2025. Srinivasan et al., DURA-CPS: A Multi-Role Orchestrator for Dependability Assurance in LLM-Enabled Cyber-Physical Systems, 2025.

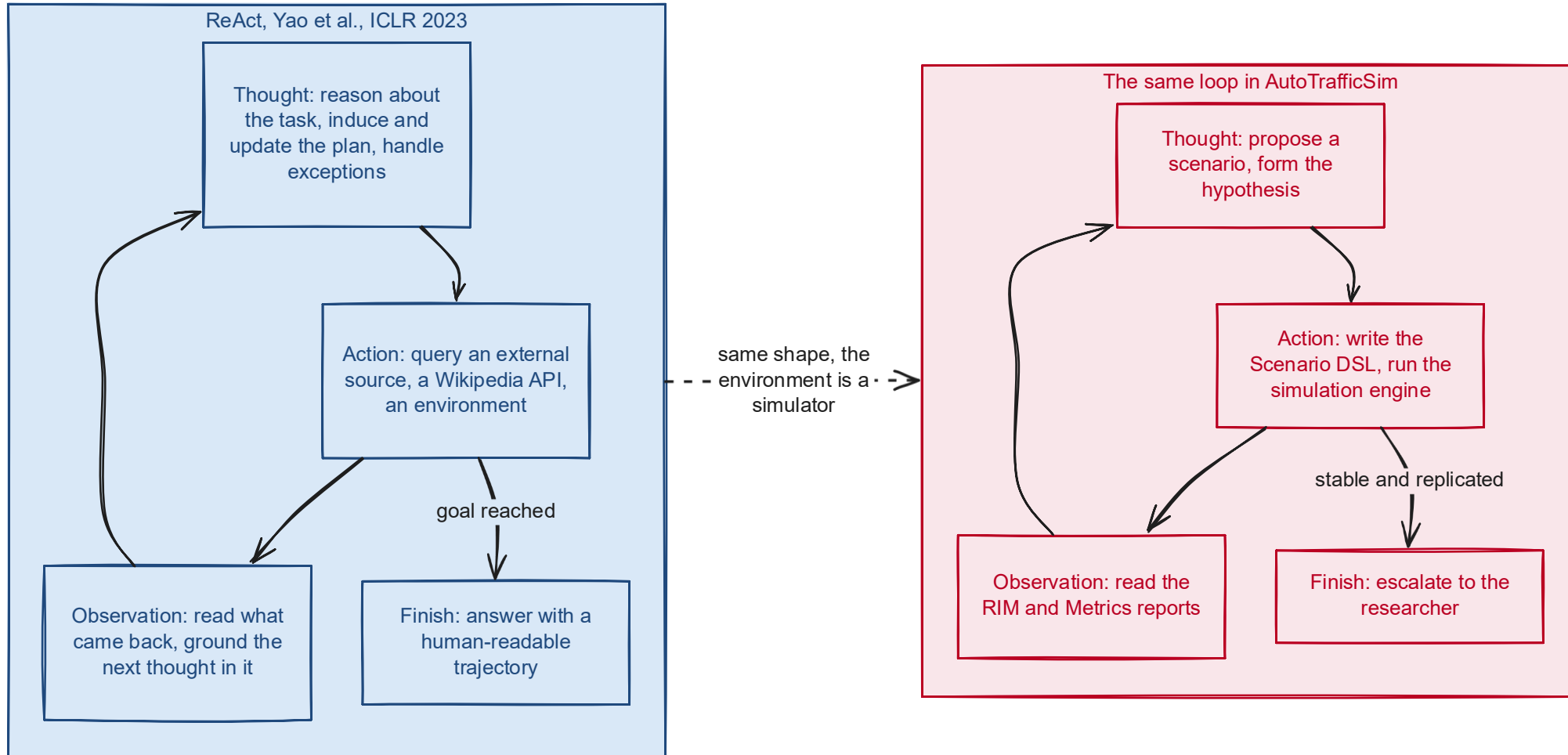
Where AutoTrafficSim agent sits in the Methods



AutoTrafficSim

- Tool-based, single-agent architecture
- External tools: Multi-tool calling harness to self complete traffic simulation task
- Tool interface: Various part of the harness architectural loops
- Iterative refinement through the Adaptive Feedback Loop
- Extendibility: Multi-Agent harness

How an agent works: reason, act, observe



“Reasoning traces help the model induce, track, and update action plans as well as handle exceptions, while actions allow it to interface with and gather additional information from external sources.” Yao et al., ReAct, ICLR 2023.

Capability is rising. Reliability is keeping up

Towards a Science of AI Agent Reliability

Stephan Rabanser Sayash Kapoor Peter Kirgis Kangheng Liu Saiteja Utpala
Arvind Narayanan
Princeton University

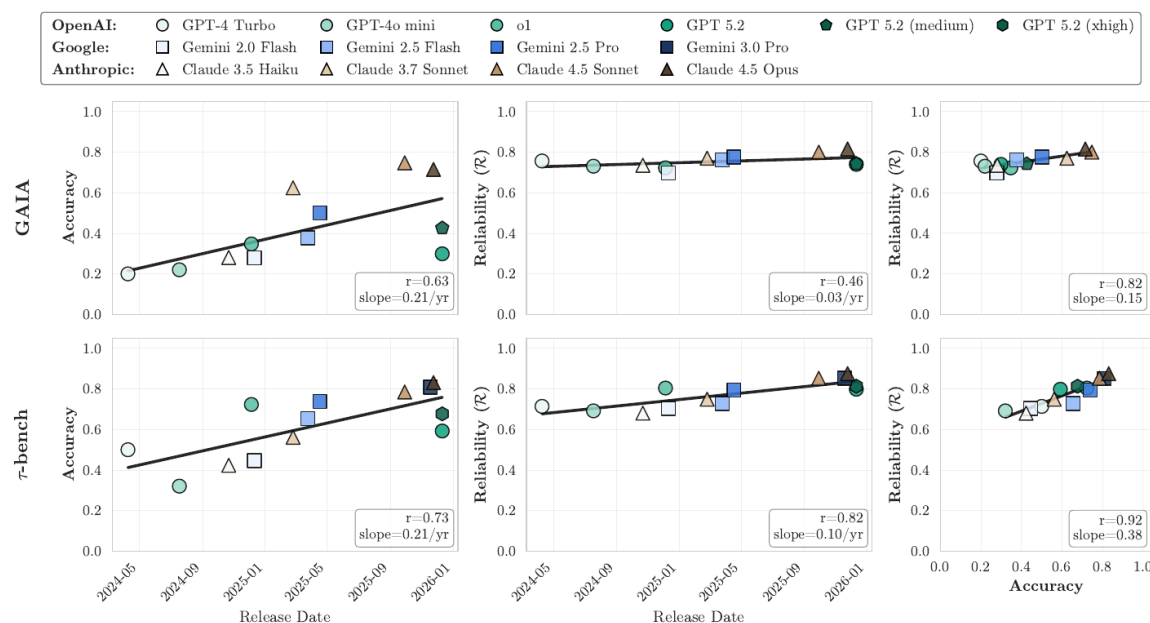


Figure 1: **Reliability gains lag behind capability progress.** Overall reliability shows slow improvement over time. While accuracy rises steadily across both benchmarks (left), reliability trails behind (center), and the relationship between the two varies across benchmarks (right), indicating that accuracy gains do not automatically yield reliability.

“Recent capability gains have only yielded small improvements in reliability.”

“Accuracy gains do not automatically yield reliability.”

Agents needs verification and validation.

Rabanser et al., Towards a Science of AI Agent Reliability. 2026.

Software engineering is already living this:

SWE vs AIE

BACKGROUND

Software Engineering (SWE)

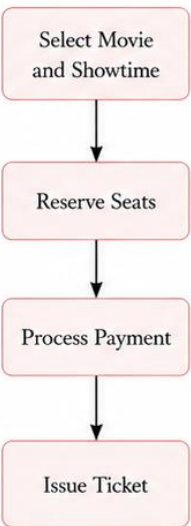
Human-directed deterministic pipeline

Human-directed task

Select a movie, choose a showtime after 7 PM, pick two available seats away from the front rows, and complete payment if the total is under \$40.

```
class TicketService:
    def bookTicket(request: BookingRequest): BookingResult =
        val selection = selectMovieAndShowtime(request) // A
        val seat      = reserveSeat(selection)           // B
        val payment   = processPayment(seat)             // C
        val ticket    = issueTicket(payment)            // D
        ticket
```

Execution Flow



- A Select** — User chooses the movie and showtime.
- B Reserve** — System checks availability and locks the selected seats.
- C Pay** — System performs a structured payment transaction.
- D Issue** — System generates and sends the ticket confirmation.

AI Engineering (AIE)

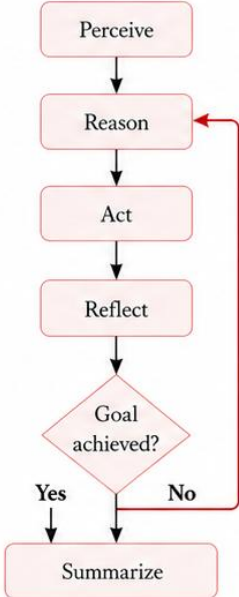
Goal-directed agentic reasoning loop

Goal-directed request

“Find two good seats for a movie tonight after 7 PM, under \$40 total, and avoid the front rows.”

```
class TicketAgent:
    def bookTicket(userRequest: String): String =
        while !isComplete() do
            val observation = perceive(userRequest) // A
            val plan        = reason(observation)   // B
            val result       = act(plan)            // C
            reflect(result) // D
        summarize()
```

Agent Loop



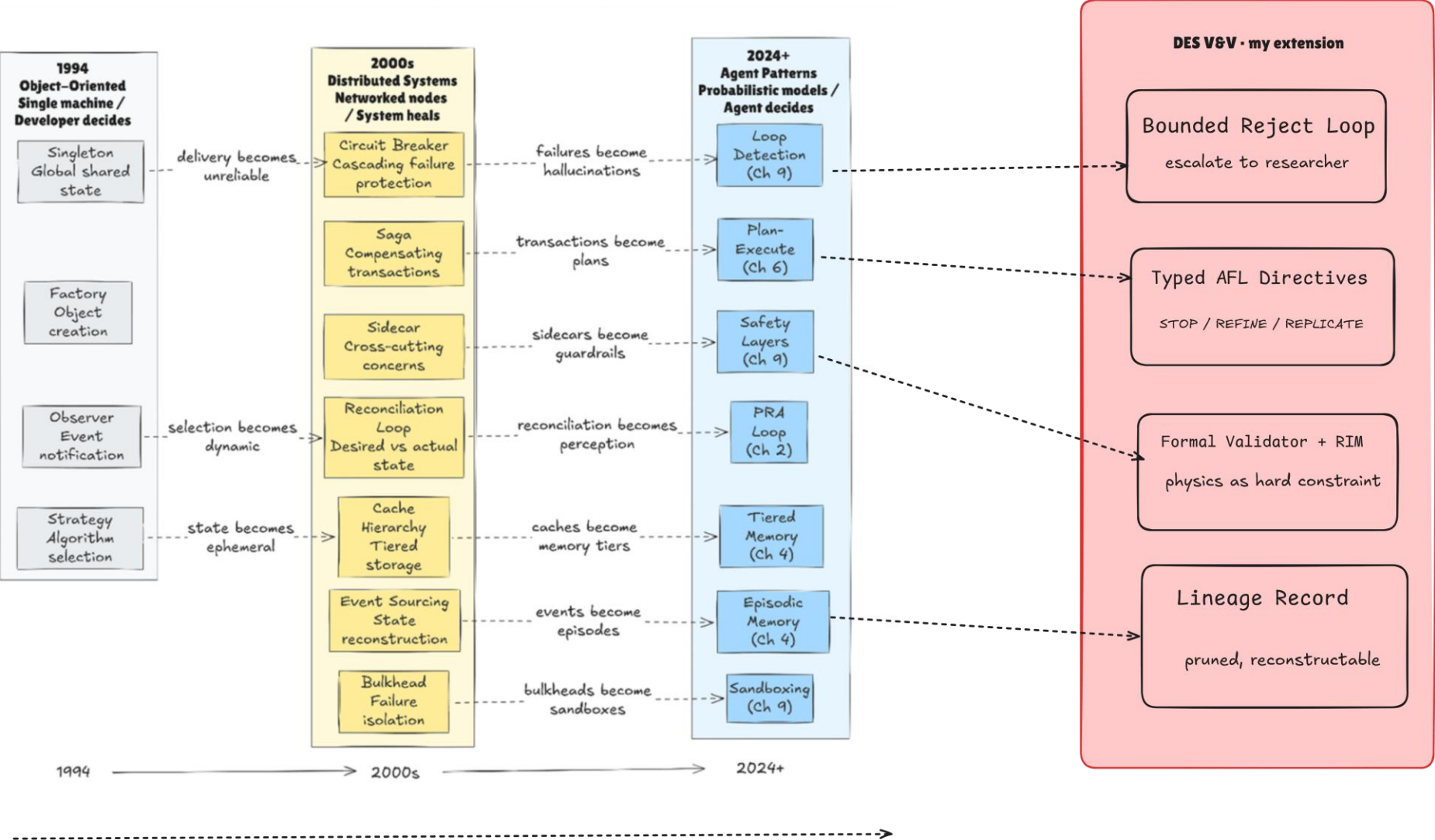
- A Perceive** — Understand the user’s constraints: tonight, after 7 PM, under \$40, and avoid the front rows.
- B Reason** — Compare theaters, movies, showtimes, prices, and seat maps.
- C Act** — Search listings, choose candidate seats, and attempt a reservation.
- D Reflect** — Check whether the seats, price, and availability satisfy the goal; loop again when necessary.

“A clear evolution based on agentic reasoning frameworks is gaining widespread recognition, gradually replacing traditional development approaches.”



Software engineering is already living this:

SWE vs AIE



“A clear evolution based on agentic reasoning frameworks is gaining widespread recognition, gradually replacing traditional development approaches.”

My Extension

Towards LLM scientific Intelligence and Discovery and Motivation.

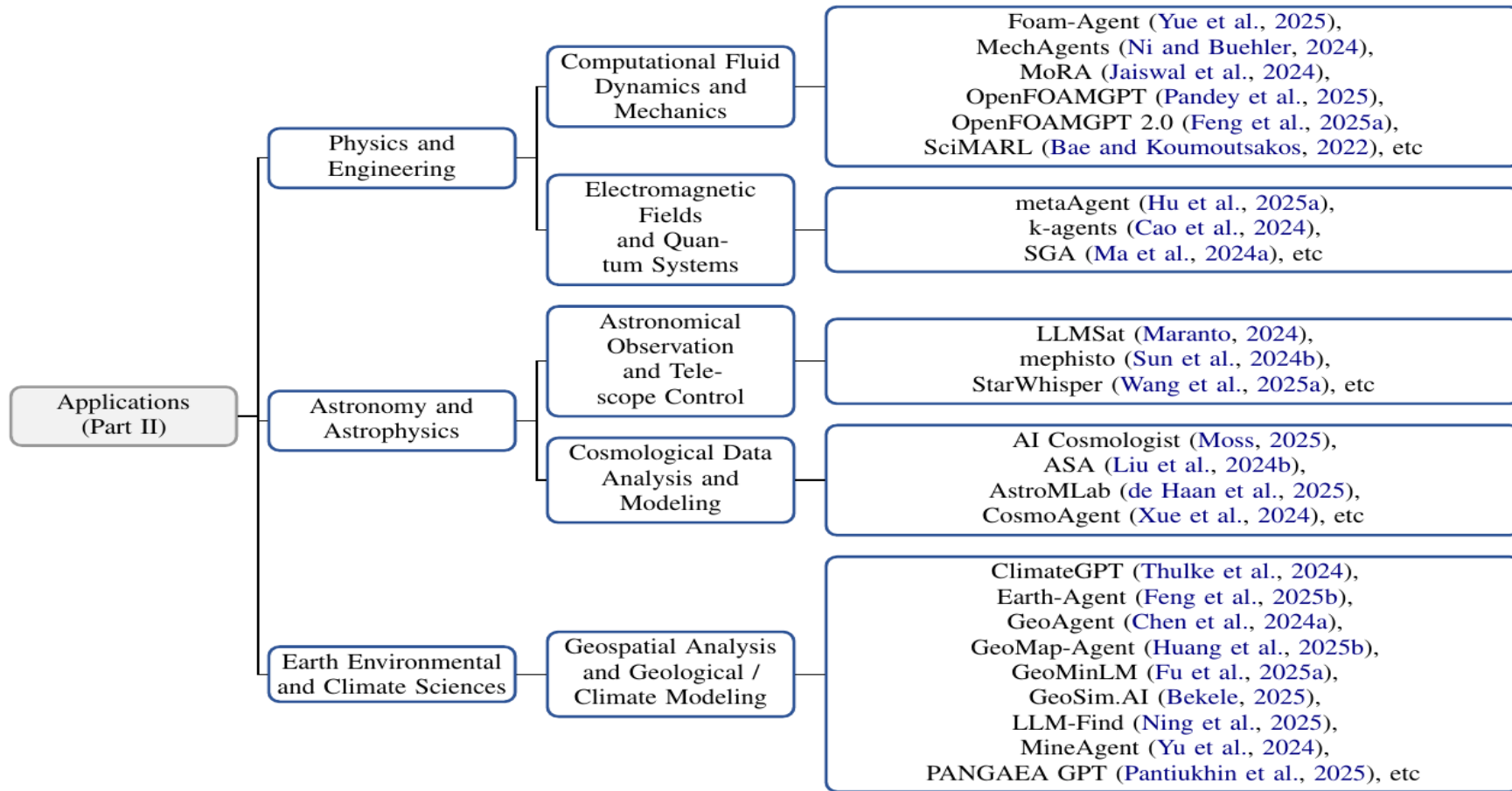


Figure 13: Applications of representative LLM-based scientific agents (Part II): Physics & Engineering; Astronomy & Astrophysics; Earth, Environmental & Climate Sciences.

Ren et al., Towards Scientific Intelligence: A Survey of LLM-based Scientific Agents, arXiv:2503.20447v3, 2025.

Traffic Simulation Across Levels of Details

MICROSCOPIC

- Every vehicle has position, speed, car-following, and lane-changing behavior that is updated at small time steps.

MACROSCOPIC

- Aggregate flow, density, and speed.
- Traffic is represented as a fluid.

MESOSCOPIC

- Individual vehicles remain identifiable,
- Movement is simplified through links, queues, or scheduled events.

THE SPLIT USED IN Q3

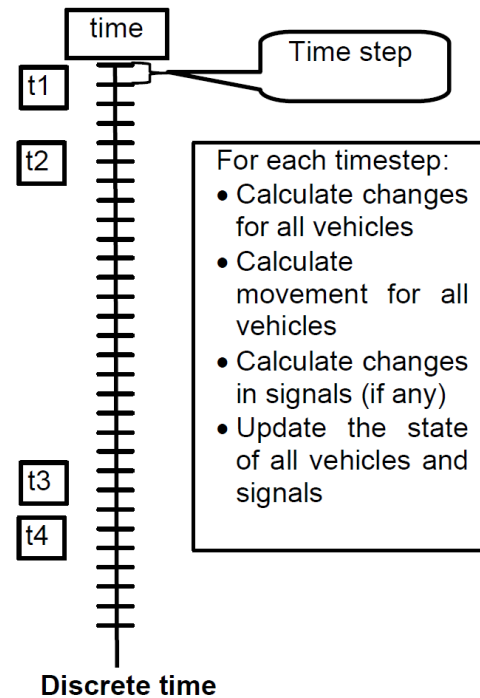
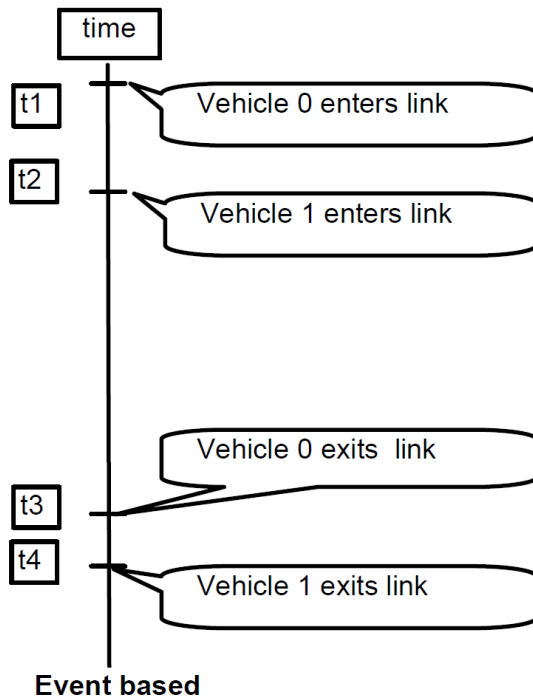
- **Event-based mesoscopic simulation**
- **Discrete-time microscopic simulation**

Burghout, Hybrid Microscopic-Mesoscopic Traffic Simulation, 2004.



School of Computing
UNIVERSITY OF GEORGIA

Director orchestrated simulation time



DIRECTOR

- The Director coordinates the simulation components and controls the simulation clock.

DISCRETE-EVENT EXECUTION

- The Director maintains time-ordered agenda.
- Selects the earliest scheduled activation and advances the clock to that event's timestamp.

DISCRETE-TIME EXECUTION

- The Director advances the model components through successive time steps and updates their state.

WHY IT MATTERS

- The same Director-component pattern supports both time representations used in this research.

Verification and Validation of Simulation

SARGENT: VALID FOR A PURPOSE

- A model is developed for a specific purpose.
- Validity is judged relative to that purpose and an acceptable range of accuracy.

BALCI: TWO DIFFERENT QUESTIONS

- Verification: Was the model implemented correctly?
- Validation: Is it the correct model for its intended use?

THE STANDING ASSUMPTION

- **Researchers define objectives**
- **Perform the checks**
- **decide what is acceptable.**

DES V&V ACTIVITIES

Conceptual-model validation

- Test assumptions and theories
- Use face validation during model development.

Computerized-model verification

- Check implementation correctness and explore model behavior during development.

Operational validation

- Compare model outputs with observed system behavior across relevant conditions.

Data validation

- Check whether the data used to build and evaluate the model are adequate and correct.

The V&V Gap AutoTrafficSim Addresses

THE ENGINEERING GAP

Simulation engines execute the model they are given. Researchers still translate scientific questions into backend code or configuration, inspect traces, diagnose failures and revise experiments manually.

THE ASSURANCE GAP

An agent can accelerate that work, but agent-generated models still need

- Physical validity gates
- Structured feedback
- Reconstruct-able lineage before the result is scientifically defensible.

THE TEST CASE

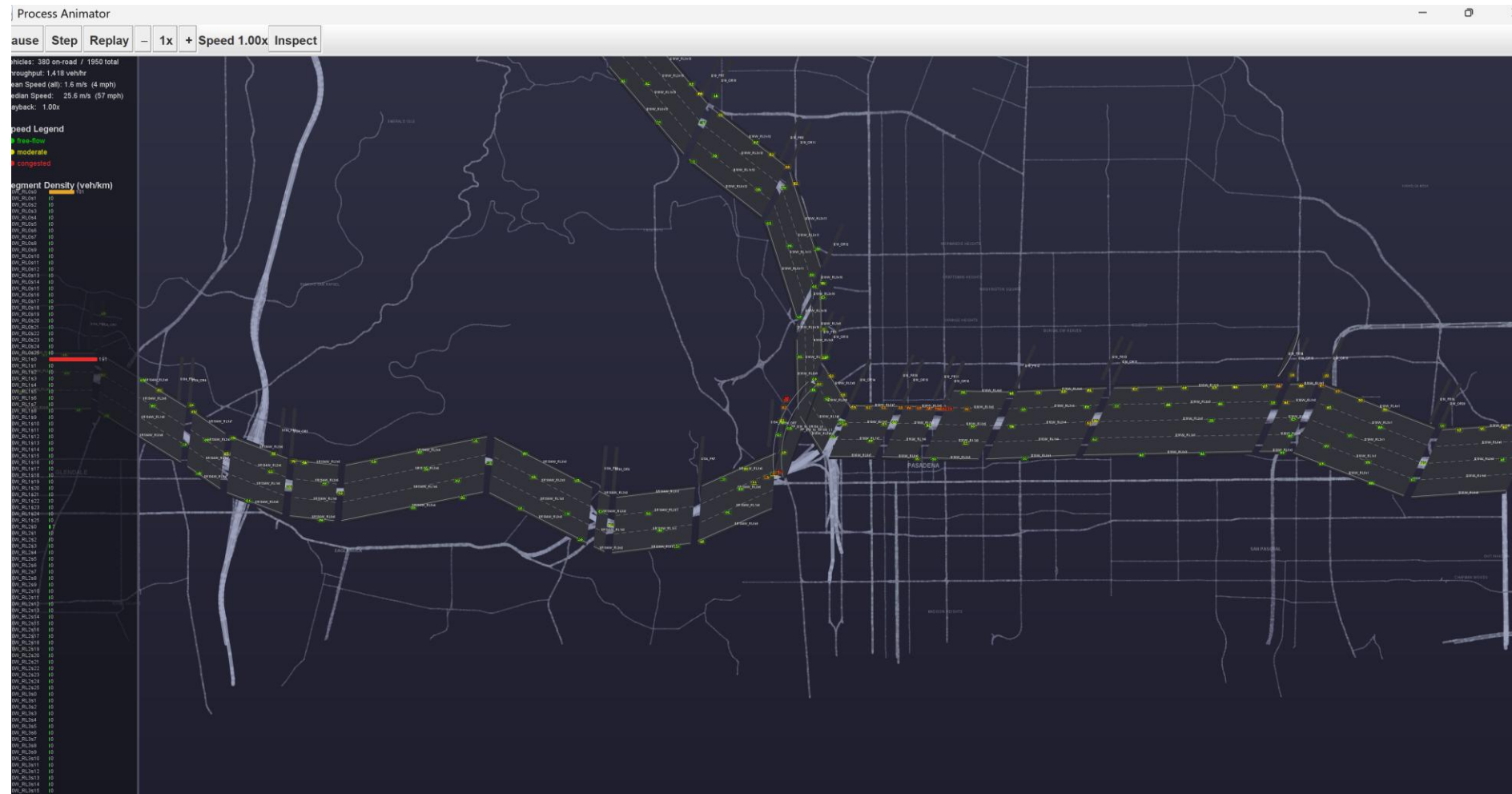
Wildfire evacuation and contraflow, where departure timing, compliance and rerouting are weakly observed but can change the policy recommendation.

- Agent can generate scenario as a fast speed than human can

Scalation Simulation Engine Animation of North LA: *Evacuation Corridor*

Case Study: North LA Freeway for Contraflow: I210 – SR134

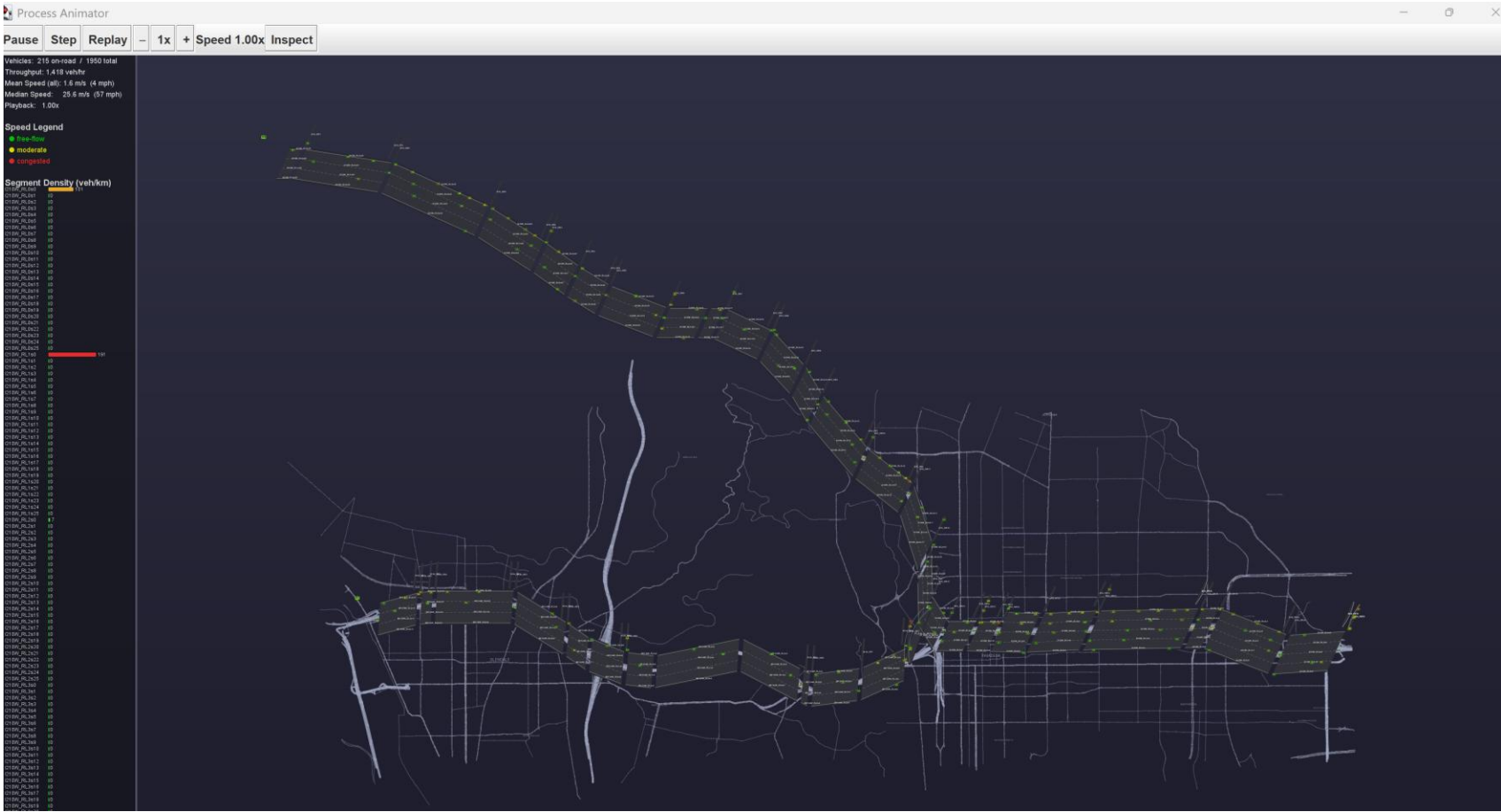
Node Counts 28,422
Links Counts 79,307
Network Length 11,596
Scale 2.8B Events



Scalation Simulation Engine Animation of North LA: *Evacuation Corridor*



Kepler.gl Map



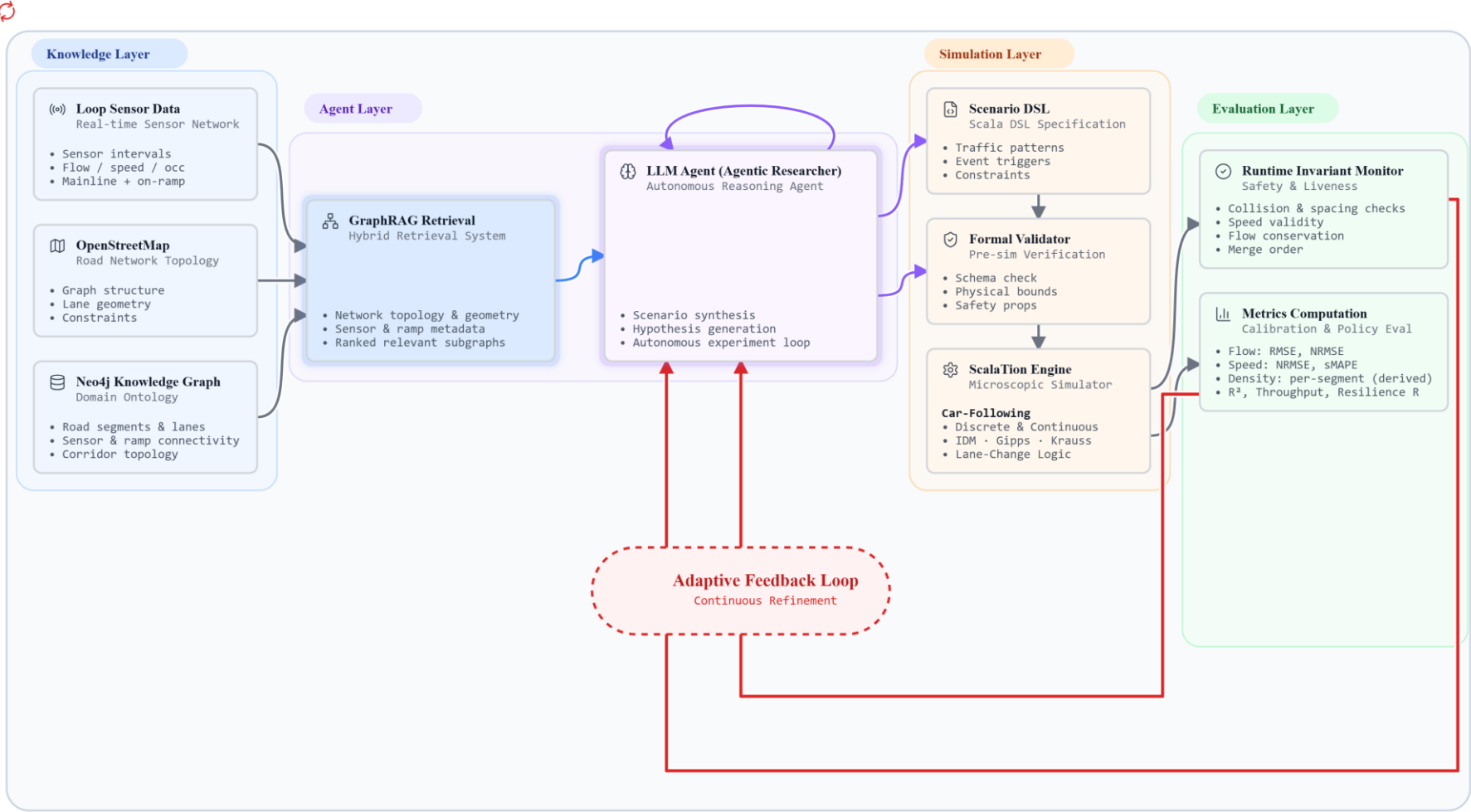
Proposed dissertation architecture: AutoTrafficSim

An Agentic Framework for Autonomous Traffic Scenario Synthesis, Verification, and Scientific Discovery

AutoTrafficSim: An Agentic Framework for Autonomous Traffic Scenario Synthesis

GraphRAG-Enhanced LLM Agent System for Microscopic Traffic Simulation

Dissertation Proposal Draft



Some Motivation Starting with Contraflow

DEFINITION

Contraflow is a temporal reversal or reallocation of inbound traffic lanes for outbound evacuation.

PURPOSE

- Use extensively during natural disaster such as hurricane
- Increase evacuation capacity away from the threatened area.
- Use existing road infrastructure during an emergency.

WHY IT MAY FAIL

- Vehicles may not reach the contraflow entrance.
- Crossover and termination points may create new bottlenecks.
- Low compliance may leave the added capacity underused.
- Downstream roads may not absorb the additional flow.
- Emergency and rescue access must remain available.

BOTTOM LINE

Contraflow is not more lanes

It's a strategy used to manage available lane resources during emergency

Traffic Simulation of LA wide fire with Contraflow

LOS ANGELES WILDFIRE MOTIVATION

On January 7, 2025, the Eaton Fire triggered mass evacuation around Altadena and Pasadena.

Severe congestion degraded performance along the I-210 / SR-134 corridor for hours.

Alternative contraflow plans could not be tested during the real emergency.

RESEARCH QUESTION

- Does contraflow improve evacuation throughput and clearance for traffic flow during wide fire
- When does it move congestion to another bottleneck?

Scenario Variables

Departure timing
Household vehicle demand
Route-guidance compliance
Dynamic rerouting
Fire and smoke conditions
Road availability

Outcomes

Evacuation throughput
Exposed vehicles
Queue and bottleneck location
Congestion-clearance time
Resilience

Take Away

Simulation provides a safe none invasive environment to counterfactually evaluate scenarios such as contraflow operations and wildfire evacuations

Verification and Validation of LLM-Generated DES Experiments

Q1 WRITTEN QUESTION

“When an LLM autonomously generates and refines discrete-event simulation experiments, part of the human researcher's traditional role as guarantor of experimental validity is transferred to the agent. Discuss the verification and validation challenges this introduces for your proposed AutoTrafficSim architecture.”

- Distinguish physically invalid traces from valid but unexpected behavior.
- Established formal, empirical, and methodological evidence for scientific soundness.
- Prevent metric overfitting in the Adaptive Feedback Loop.
- Differentiate AutoTrafficSim from automated scenario generation.

Q1 FOLLOW-UP

- When does the feedback loop stop?
- How is quarantine specified statistically?
- How can a researcher navigate 1,000 iterations?
- What does the agent decide, and what remains human?
- How is scenario-state growth pruned?

Q1.1 Verification and Validation Challenges to AutoTafficSim:

Classic DES V&V ACTIVITIES : Deterministic

Conceptual-model validation

- Making sure the model works for simple scenarios

Computerized-model verification

- Check implementation correctness and explore model behavior during development.

Operational validation

- Compare model outputs with observed system behavior across relevant conditions.

Data validation

- Check whether the data used to build and evaluate the model are adequate and correct.

Agentic V&V Challenges: Probabilistic

Stochasticity and unpredictability

- The same prompt may produce different scenario proposals.

Hallucinations

- Wrong context may enter through the Knowledge Layer or Scenario DSL.

Emergent unintended behavior

- A valid DSL may still produce a physically invalid or misleading trace.

New pipeline vulnerabilities

- An error can propagate from scenario to simulation, metric, and feedback, then be reinforced.

Sargent, An Introduction to Verification and Validation of Simulation Models, 2013.

Balci, Verification, Validation, and Testing, 1998.

Koohestani, AgentGuard: Runtime Verification of AI Agents, 2025.

Runtime Assurance and Verification: AgentGuard

Dynamic Probabilistic Assurance via Markov Decision Process

AgentGuard: Runtime Verification of AI Agents

Roham Koohestani
JetBrains Research, The Netherlands
roham.koohestani@jetbrains.com

Abstract—The rapid evolution to autonomous, agentic AI systems introduces significant risks due to their inherent unpredictability and emergent behaviors; this also renders traditional verification methods inadequate and necessitates a shift towards probabilistic guarantees where the question is no longer if a system will fail, but the probability of its failure within given constraints.

(S, A, P, R, γ) where S is a finite set of states, A is a finite set of actions, $P(s'|s, a)$ is a probability of transitioning to state s' after taking action a in state s , $R(s, a, s')$ is the immediate reward received after transitioning from s to s' via action a , and γ is a discount factor that determines the importance of

Observe:

Capture agent I/O, tool calls, and outcomes.

Abstract:

Convert the trace into state transitions:

$$s \rightarrow a \rightarrow s'$$

Learn:

- Continuously update the Markov Decision Process and its transition probabilities.

Verify:

Use probabilistic model checking against predefined PCTL properties.

Respond:

Report quantitative guarantees and trigger alerts, automated responses, or human intervention when a threshold is breached.

Runtime Invariant Families

$$x_j(t) - x_i(t) - \ell_i > 0, \quad 0 \leq v_i(t) \leq v_{\max}, \quad -b_{\max} \leq a_i(t) \leq a_{\max}$$

where

- x = rear-bumper position
- ℓ_i = the follower's length
- j, i = every leader j and follower i
- $v_{\max}$ = maximum speed
- $a_{\max}, b_{\max}$ = acceleration and braking limits

Flow conservation: Vehicles entering a segment equal vehicles exiting plus those remaining inside.

Collision exclusion: Every leader–follower pair maintains positive physical spacing.

Kinematic bounds: Vehicle speed, acceleration, and deceleration remain within admissible limits.

Lane integrity: Vehicles do not cross non-adjacent lanes, and merges preserve queue order.

Event ordering: ScalaTion orchestrates in timestamp order through its Director agenda.

Q1.2: Reject, Quarantine, or Escalate

Q1.2

THREE-WAY VERDICT

REJECT

A hard physical or computational invariant fails flagged by the RIM.

QUARANTINE

The invariants hold, but the behavior disappears under different seeds, smaller time steps, or changed boundary conditions.

ESCALATE

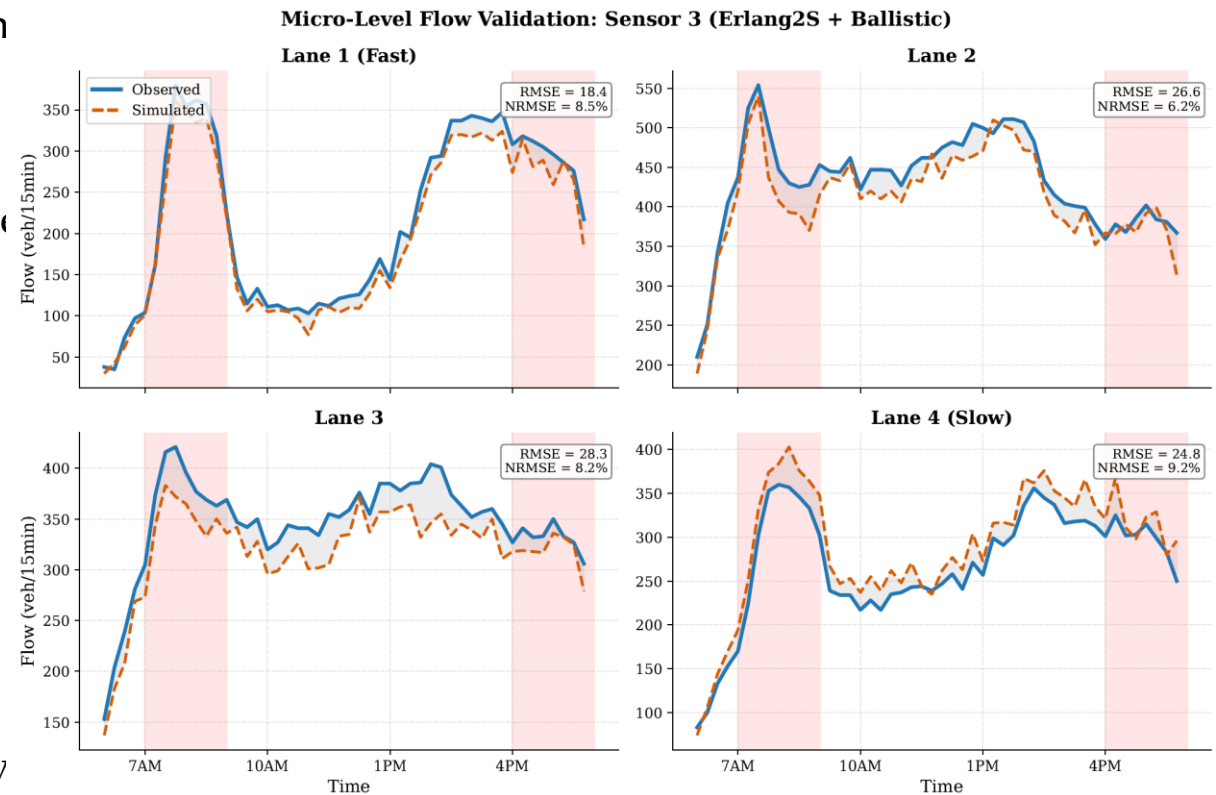
The invariants hold and the behavior survives those checks. The researcher determines whether it is a discovery, artifact, or model defect.

BOTTOM LINE

Unexpected $\neq$ invalid.

Invariant-preserving and stable behavior may be scientifically valuable.

Bishi, Bowman, and Miller



Q1.2: Where Each Decision Is Made

FORMAL VALIDATOR — BEFORE EXECUTION

- Malformed Scenario DSL
- Density above jam density
- Invalid arrival schedule
- Lane assignment inconsistent with topology

RUNTIME INVARIANT MONITOR — DURING EXECUTION

- Flow conservation
- Collision exclusion
- Kinematic bounds
- Lane integrity
- Event ordering

ADAPTIVE FEEDBACK LOOP — AFTER EVALUATION

- Returns structured diagnostics to the agent
- Supports scenario revision
- Does not declare scientific discovery

Agentic Objective and Termination Rule

Multiple metrics, hard invariants, and robustness checks guard against metric gaming.

$$\theta^* = \arg \min_{\theta} \sum_i w_i M_i(\theta)$$

subject to $\text{RIM}(\theta) = \text{PASS}$

where

θ = scenario and model parameters

$M_i(\theta)$ = normalized error per metric: M_1 NRMSE flow, M_2 NRMSE speed, M_3 NRMSE occupancy, ...

w_i = metric weights, pre-set by the researcher

$\text{RIM}(\theta)$ = Runtime Invariant Monitor verdict: a hard constraint, so invalid regions of the search space are unreachable, not merely penalized

my proposal: objective and stopping criterion (Q1.4)

EXPLICIT TERMINATION LOGIC

- $\text{RIM} = \text{FAIL}$ -- Reject or quarantine the candidate.
- $\text{RIM} = \text{PASS}$ and $J(\theta) > J^*$ --- Refine θ .
- $\text{RIM} = \text{PASS}$ and $J(\theta) \leq J^*$ --- Objective reached; proceed to independent replication.
- Budget exhausted before reaching the objective --- stop as inconclusive.

$$\bar{\bar{w}} = \frac{1}{n_r} \sum_{i=0}^{n_r-1} \bar{w}_i \quad \widehat{\sigma}_{\bar{w}} = \sqrt{\frac{1}{n_r - 1} \sum_{i=0}^{n_r-1} (\bar{w}_i - \bar{\bar{w}})^2}$$

$$\text{ihw} = t_{n_r-1, 1-\alpha/2} \frac{\widehat{\sigma}_{\bar{w}}}{\sqrt{n_r}} \quad \text{CI}_{1-\alpha} = [\bar{\bar{w}} - \text{ihw}, \bar{\bar{w}} + \text{ihw}]$$

$$\begin{cases} \text{ihw} > \varepsilon & \implies \text{continue replicating} \\ \text{ihw} \leq \varepsilon & \implies \text{stop: precision reached} \\ n_r = N_{\max} \wedge \text{ihw} > \varepsilon & \implies \text{inconclusive} \end{cases}$$

where

n_r	= number of independent replications (start $n_r = 10$)
$\bar{w}_i$	= output mean from replication i
$\bar{\bar{w}}$	= grand mean of the replication means
$\widehat{\sigma}_{\bar{w}}$	= sample standard deviation of the replication means
$t_{n_r-1, 1-\alpha/2}$	= Student- t critical value, $n_r - 1$ degrees of freedom
ihw	= interval half-width: the precision achieved so far
$\text{CI}_{1-\alpha}$	= the $1 - \alpha$ confidence interval on the grand mean
ε	= maximum acceptable interval half-width (the precision target)
$N_{\max}$	= maximum replication budget

COMMITTEE: How many replications, and what statistical test and “Good enough”?

Method of Independent Replication (MIR):

Appropriate for finite-duration evacuation experiments

- Run the complete simulation independently n_r times.
- Use a different random-number stream for every replication.
- Record one output mean from each replication.
- Calculate a Student- t confidence interval across the run means.

Q1.3 Establishing formal, Empirical and Methodical Evidence

Q1.3

FORMAL EVIDENCE

- DSL contracts and schema checks
- Formal Validator checks
- Runtime invariants and admissibility logs

EMPIRICAL EVIDENCE

- Traffic: flow, speed, occupancy, and travel time
- Evacuation: clearance, queues, exposure, and bottlenecks
- Sensitivity across the scenario matrix

METHODOLOGICAL EVIDENCE

- Hypothesis and parent scenario
- Seeds, configurations, and DSL/code versions
- Metrics and decisions sufficient to reconstruct the run

Sargent, An Introduction to Verification and Validation of Simulation Models, 2013.
Rabanser et al., Towards a Science of AI Agent Reliability, 2026.

Pruning and Lineage at Scale

Q1.4

COMMITTEE: After 1,000 iterations, how is the trail navigated, and scenario growth controlled?

LINEAGE INDEX

- Scenario and parent IDs
- Hypothesis, reason codes, seeds, and metrics
- Append-only metadata retained for every branch

ACTIVE PRUNING

- Hard prune Formal Validator or RIM failures
- Keep at most K admissibles
- K is fixed by the researcher before the search

RESEARCHER VIEW

- Filter by verdict, reason code, or metric change
- Navigate ancestors and descendants
- Review escalations and representative samples

Active runs:

Keep the full traces for scenarios the agent is still investigating.

Escalated runs:

Keep the full traces for unexpected results sent to the researcher for scientific interpretation.

Representative runs:

Keep selected full traces that summarize common outcomes, parameter regions, failures, or successful scenario families.

Q1.5. Differentiate: AutotrafficSim versus Scenario Generators

Q1.5

SCENARIO TOOLS: Deterministic

A researcher defines the scenario space and parameter ranges.

The researcher specifies:

- Network and evacuation demand
- Contraflow configuration
- Departure and compliance assumptions
- Route-choice behavior
- Fire and road-closure conditions

The tool generates and executes the requested scenario.

The researcher decides what scenario to run next.

AUTOTRAFFICSIM: Probabilistics

The researcher defines the scientific question:

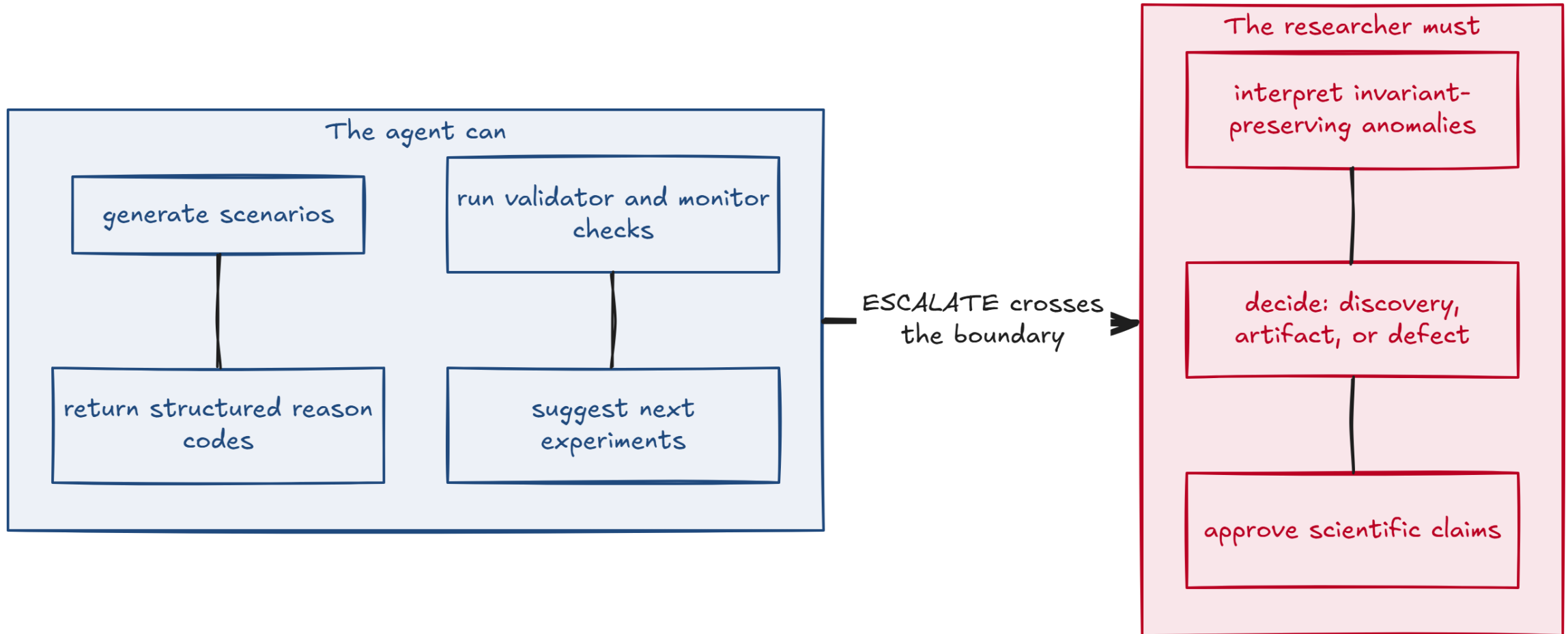
“Under which behavioral and fire conditions does contraflow on the I-210/SR-134 evacuation network reduce clearance time without increasing vehicle exposure or transferring congestion downstream?”

The agent proposes the next bounded experiment:

- Baseline: no contraflow
- Treatment: selected contraflow configuration
- Select the model to use by default
- Vary departure timing, compliance, rerouting and fire progression within researcher-approved ranges
- Execute independent replications
- Check physical invariants
- Compare clearance time, exposure, queues and bottlenecks
- Use the evidence to select the next experiment

The Human – Agent Decision Boundary

COMMITTEE: What does the agent decide, and what remains human?



Q2 Written and Follow-up Questions

Q2 WRITTEN QUESTION

- How do you model and justify behavioral assumptions such as compliance, rerouting, and panic effects that are rarely observed in standard datasets?
- What is the fundamental need for an agentic simulation framework when SUMO with calibration and optimization can already reproduce observed data?
- How does the approach go beyond fitting to enable new, generalizable knowledge?

MY RESPONSE

Q2.1 Model unobserved behaviors as explicit travel-demand, destination-choice, route-choice, and compliance inputs.

Q2.2 Keep SUMO or ScalaTion as the simulation engine and place AutoTrafficSim above it as the agentic experimentation and assurance layer.

Q2.3 Go beyond one calibrated result by identifying which assumptions change the evacuation decision and where the conclusion stops holding.

Q2 COMMITTEE FOLLOW-UP

What concrete behavioral ranges will be used?	Addressed in Q2.1
How is credibility judged when direct observations are limited?	Addressed in Q2.1
How will AutoTrafficSim be benchmarked against a standard SUMO workflow?	Addressed in Q2.2

Q2.1 Behavioral Decisions For Evacuation

BEHAVIORAL CHOICE	PEL'S MODEL COMPONENT	AUTOTRAFFICSIM IMPLEMENTATION
Whether and when to evacuate	Dynamic travel demand	A repeated binary-logit, Faster departure may worsen congestion if demand enters too quickly
Where households go	Dynamic trip distribution	Origin-destination matrix, with distance-based destination allocations where observations are limited. Agent may vary the allocation with contraflow response
Route and whether to switch	Dynamic traffic assignment	• Pre-trip: fixed route. • En-route: Responds continuously to traffic. • Hybrid: begins pre-trip and listen to radio
Response to route guidance	Assignment input	Responsiveness to guidance. Zhao and Wong used 15% rerouting as a baseline, with 0%, 50%, and 100% alternatives.
Vehicles per household	Travel-demand input	A discrete vehicle-count distribution. Zhao and Wong used 44% one vehicle, 43% two, and 13% three, approximately 1.7 vehicles per household.

Concrete Behavioral Ranges

Add concrete parameter ranges or distributions for key behaviors.

Household vehicles:

- 41–45% of evacuees used two vehicles
- 9–17% used three or more.
- These are post-wildfire survey observations from (Zhao and Wong).

Departure timing:

- Weibull or sigmoid response families may be tested as:
 - Early(10-20min) , middle (30-50min), and late departure regimes(1hour +). (Pel et al)

Guidance compliance: $p_c \in \{0.25, 0.50, 0.75, 1.00\}$.

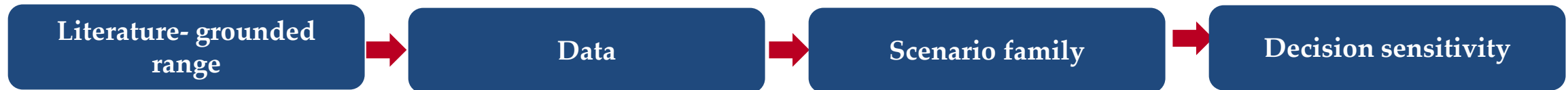
- This is an initial sensitivity grid, not a claim about the true compliance rate.

Rerouting:

- Switching requires a perceived improvement $\tau \in \{5\%, 15\%, 30\%\}$.
- These are initial experimental settings I intend to refine with local evidence.

Credibility with Limited Behavioral Ranges

Clarify how the model is judged credible when direct behavioral observations are limited.



Ground what is observed:

Use post-wildfire surveys for household vehicle use, warning response, destinations, and reported rerouting where available.

Check what traffic data can support:

Validate ordinary traffic behavior against flow, speed, occupancy, and travel time data
Enforce physical invariants on every trace.

Stress-test what is not observed:

Compliance running

Departure shape

Switching thresholds across documented scenario ranges

Q2.2 Why AutotrafficSim, Not only SUMO or ScalaTion

Follow up Question: Since SUMO can already do these tasks, the framework seems to contribute efficiency rather than new functions. Validate by benchmarking against a standard SUMO workflow.

$$\min_{\theta} \sum_t \|y_t^{\text{obs}} - y_t^{\text{sim}}(\theta)\|^2$$

where

y_t^{obs} = observed measurement at time t

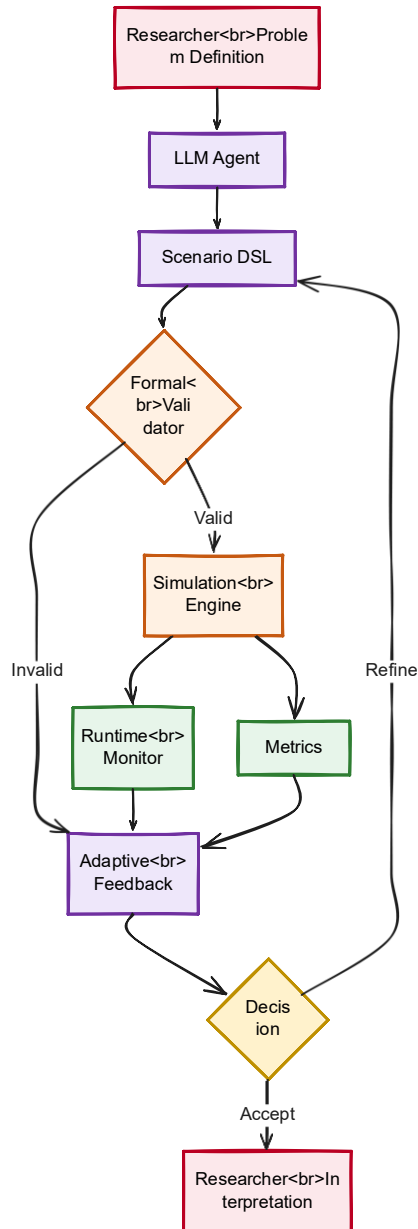
$y_t^{\text{sim}}(\theta)$ = simulated measurement under parameters θ

θ = parameters, tuned once the model structure is chosen

standard calibration formulation



Follow up Question + Q3: *Sumo can do this; this is efficiency vs function: Sumo is an Engine AutotrafficSim is Agentic; Fitting and new Knowledge*



SCENARIO 1: EXISTING ENGINE MODEL

- Researcher supplies the question, model, data, assumptions, and constraints.
- AutoTrafficSim configures the existing model through the Scenario DSL.
- Formal Validator checks the specification before execution.
- SUMO or ScalaTion executes the experiment.
- Runtime Invariant Monitor and Metrics Computation evaluate the result.
- The agent refines the experiment from the returned evidence.

SCENARIO 2: NEW MODEL: PROPOSED EXTENSION

- Researcher supplies the published equations and assumptions.
- AutoTrafficSim produces a candidate backend implementation.
- The candidate is compiled and tested in a sandbox or temporary clone.
- Tests cover numerical behavior, physical invariants, and empirical evidence.
- Researcher reviews the mathematics, code, tests, and assumptions
- Only a human-approved model is promoted

where

$Y = f(v_f, \theta_T, r, u, \phi, h)$

Y = evacuation outcome (clearance time, exposed vehicles, queue duration, bottleneck location)
 v_f = fire spread speed
 θ_T = departure-timing control
 r = route-choice regime
 u = contraflow plan
 ϕ = phasing policy
 h = household vehicle demand

Factor	Low or baseline	Middle	High or severe
Fire spread	Slow spread	Medium spread	Fast spread
Departure timing	Early departure	Middle departure	Late departure
Route choice	Pre-selected route	Dynamic rerouting	Hybrid route choice with pre-trip plan and en-route switching
Contraflow	No contraflow	Partial contraflow	Full or extended contraflow
Phased evacuation	No phasing	Phased zones	Full-area evacuation
Vehicle demand	One vehicle per household	Mixed vehicle use	High multi-vehicle demand, including towing

my submission, Q2.3: AutoTrafficSim scenario matrix for the I-210/SR-134 evacuation study

Q3 · Written and Follow-up Questions

Q3

Q3 WRITTEN QUESTION

“Review and critique the research literature on hybrid meso-micro traffic simulation.

Explain in detail how meso-scale models work and how they can be integrated with micro-scale models for effective city-wide traffic simulation.

As meso-scale models are discrete-event and micro-scale models are discrete-time, how can they be synced up. Your answer should explain the situation all the way down to mathematical equations.”

MY RESPONSE

- Distinguished cell-based CTM from event-based Mezzo.
- Used Wilco Burghout's interface requirements, loading rules, and equations.
- Treated each micro time step as an external mesoscopic event.
- Used Mobiliti and Time Warp as the PDES precedent, scoped first to North Los Angeles.

Q3 COMMITTEE FOLLOW-UP

- How does the meso–micro interface work?
- How would this be implemented in North Los Angeles?
- Is PDES fundamental or mainly required for computational scale?

Q3. Review: Burghout Foundation, Hybrid Simulation Is a Consistency Problem

Burghout's Consistent requirements

- Network and route-choice consistency
- Boundary traffic dynamics
- Traffic-performance consistency
- Communication transparency
- Workable architecture.

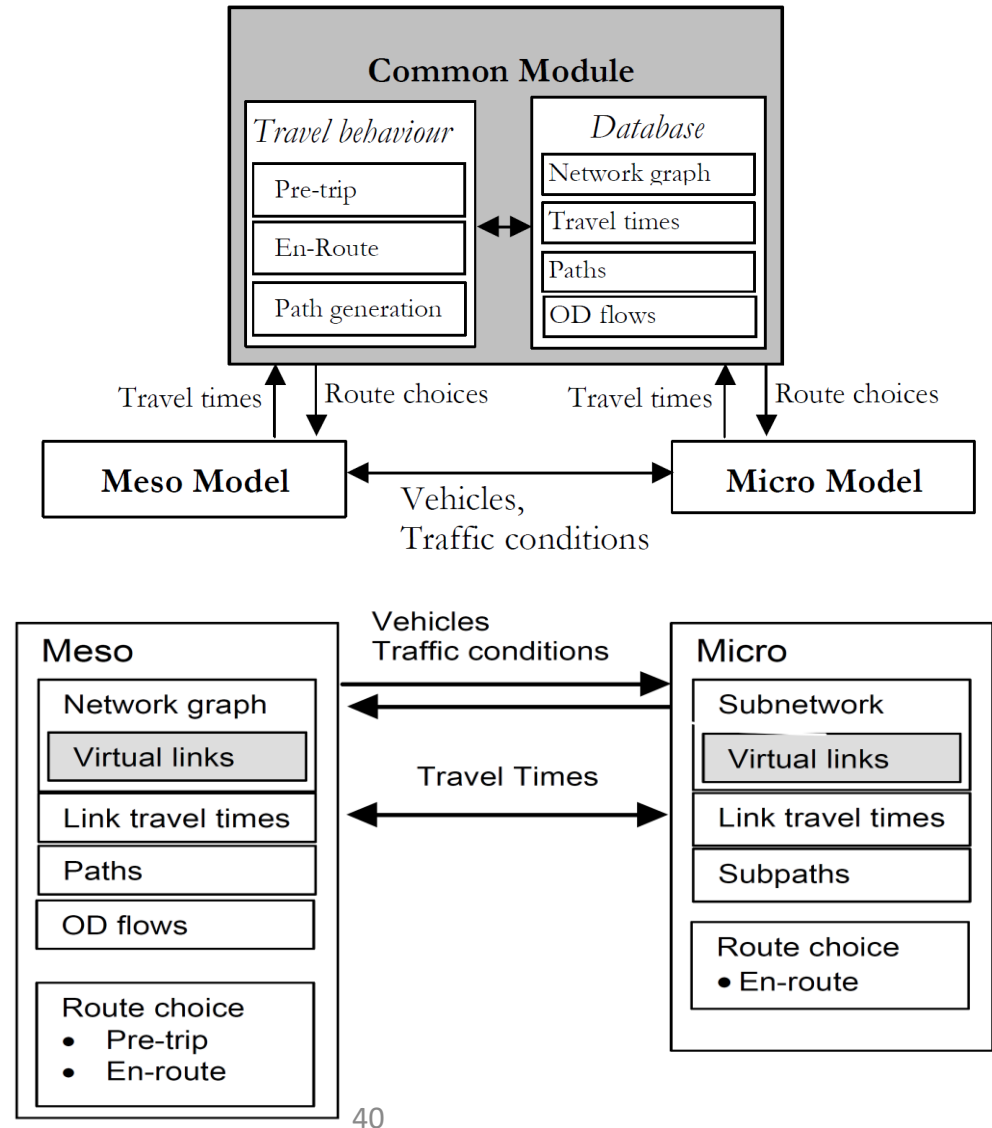
Similarity condition

where either model is valid, both should produce nearly the same result at the mesoscopic level of aggregation.

LIMITATION

MiMe was evaluated on the Brunnsviken network.
Traffic-signal operation was simplified.

Burghout, Hybrid Microscopic-Mesosopic Traffic Simulation, 2004.



Q3. What Newer Papers Added

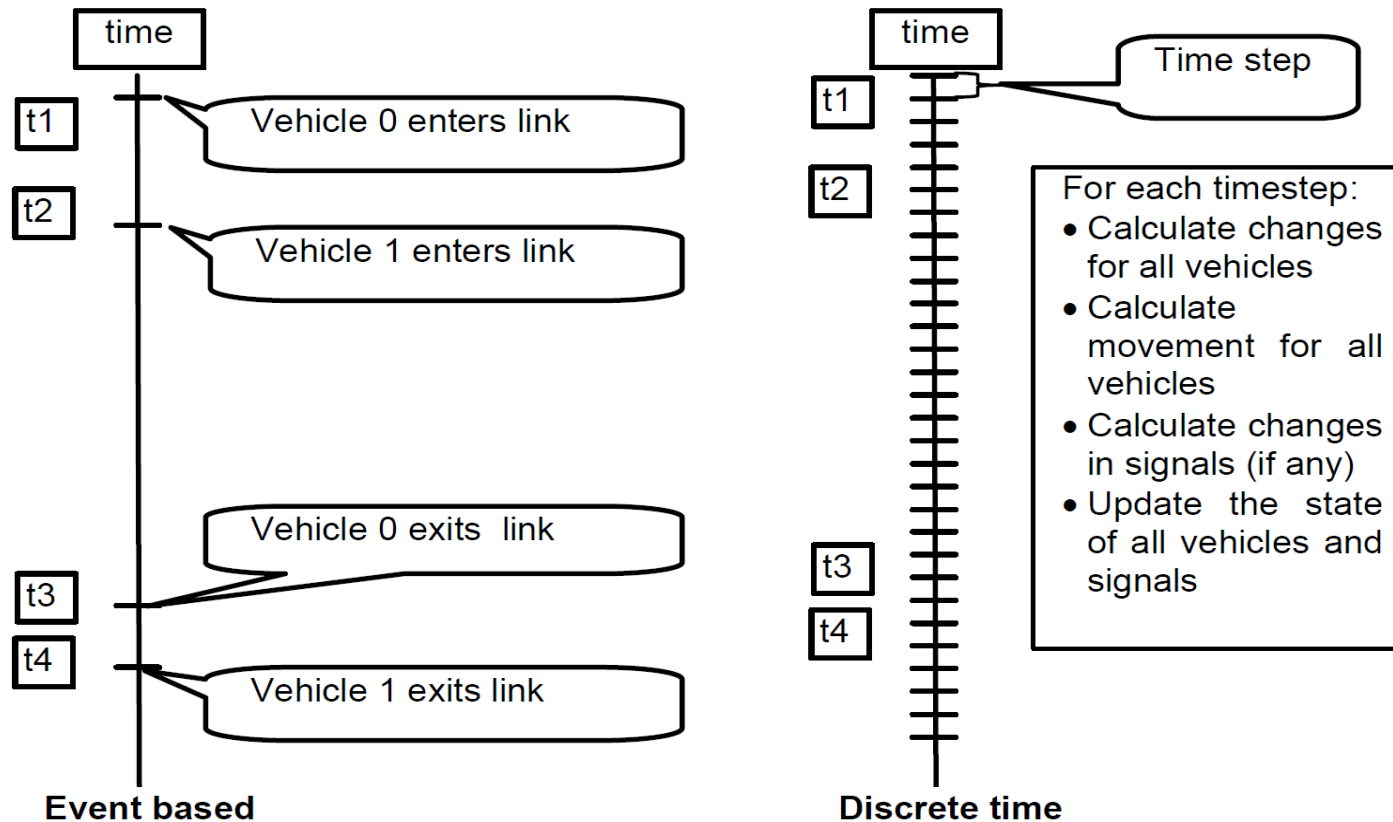
PAPER	WHAT IT ADDS	WHAT REMAINS
OTM Gomes (2022)	Interface conservation	Simple linear network
hybridPY Schweizer et al. (2024)	MATSim/BEAM - SUMO workflow	Boundary dynamics not validated
Hybrid large-scale simulation Schuhmann et al. (2025)	Flow and speed validation	Loading, blocking, and queue transfer not tested
POLARIS de Souza et al. (2019)	Richer mesoscopic model with traceable travelers	High-fidelity links left as future work
Differentiable hybrid Son et al. (2022)	Gradient flow across model representations	Traffic-state consistency not established

- *The literature advances interfaces, toolchains, validation, and optimization.*
- *No paper in this review closes the full city-scale boundary-consistency problem.*

*de Souza et al. (2019); Son et al. (2022); Gomes (2022);
Schweizer et al. (2024); Schuhmann et al. (2025)*

Mesoscopic traffic models do not share one clock

Q3.1



Cell-based CTM Discrete-time state updates over cells at clock ticks.

Event-based Mezzo Identifiable vehicles scheduled through a future event list.

For synchronization I use Burghout/Mezzo because it is the event-based mesoscopic case named in Q3.

Daganzo, The Cell Transmission Model, 1994; Burghout et al., A Discrete-Event Mesoscopic Traffic Simulation Model for Hybrid Traffic Simulation, 2006.

Q3.2 Mezzo Speed-Density Function

- A combination of macroscopic Greenshield with a Microscopic speed density model
- Different speed assigned to vehicle due to varying link condition.
- Simple enough to represent vehicles at both model surfaces

$$V(k) = \begin{cases} V_{\text{free}}, & k < k_{\text{min}}, \\ V_{\text{min}} + (V_{\text{free}} - V_{\text{min}}) \left[1 - \left(\frac{k - k_{\text{min}}}{k_{\text{max}} - k_{\text{min}}} \right)^a \right]^b, & k_{\text{min}} \leq k \leq k_{\text{max}}, \\ V_{\text{min}}, & k > k_{\text{max}} \end{cases}$$

where

$V(k)$ = speed assigned to the entering vehicle

k = density on the running part of the link

V_{free} = free-flow speed

V_{min} = minimum speed

$k_{\text{min}}, k_{\text{max}}$ = density thresholds

a, b = model parameters

Q3.2 Mezzo: Shockwave Equation

Helps the meso-model to compute the speed at which congestion propagates through the link (network)

As traffic Jam propagates like wave, this helps to compute how fast that wave moves

$$w_{AB} = \frac{q_A - q_B}{k_A - k_B}$$

where

w_{AB} = shockwave speed

q_A, q_B = flows upstream and downstream of the node

k_A, k_B = densities upstream and downstream of the node

LWR – Vehicle Conservation Law

LWR Law must hold when vehicle entered and leaves a meso-micro boundary

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho V)}{\partial x} = \frac{\partial \rho}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

where

ρ = density

V = speed

$Q = \rho V$ = flow

Lighthill & Whitham (1955); Richards (1956)

Microscopic Gipps-Style Update

$$v_{\text{safe}}(s, v_l) = -b\Delta t + \sqrt{b^2\Delta t^2 + v_l^2 + 2b(s - s_0)}$$

$$v(t + \Delta t) = \min \{v(t) + a\Delta t, v_0, v_{\text{safe}}(s, v_l)\}$$

where

s = gap to the leader

s_0 = minimum gap

v_l = the leader's speed

a = maximum acceleration

b = braking rate

v_0 = desired speed

Δt = microscopic time step

Gipps-style safe-speed rule

Microscopic Models includes how single vehicles affects traffic.

Involves the relationship between A car follower with his leader directly (leader or ego car)

Virtual Vehicle across the boundary

Q3.2

$$x_{\text{virtual}}(t_{\text{current}}) = x_{\text{end of micro link}} + v_{\text{virtual}} (t_{\text{current}} - t_{\text{exit}})$$

where

t_{current} = current simulation time

t_{exit} = time the last vehicle on that lane exited the micro region

v_{virtual} = speed assigned to that vehicle in the mesoscopic model

Burghout (2004); Burghout et al. (2005)

Boundary Loading of Vehicle Create Artificial Waves

$$V = \begin{cases} V_{\text{front}}, & t_1 < t_h \leq t_2, \\ \alpha V_{\text{desired}} + (1 - \alpha)V_{\text{front}}, & \alpha = \frac{t_h - t_2}{t_3 - t_2}, \quad t_2 < t_h \leq t_3, \\ V_{\text{desired}}, & t_h > t_3 \end{cases}$$

where

V = assigned initial speed

V_{front} = speed of the leading vehicle in the same lane

V_{desired} = the entering vehicle's desired speed

t_h = time headway to the leading vehicle

t_1, t_2, t_3 = thresholds calibrated from field data ($t_h \leq t_1$: do not load)

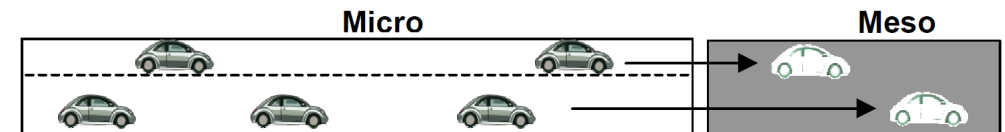
Burghout et al. (2006), multi-regime loading

$V = V_{\text{front}}$ when the entering vehicle is close to its leader,

$V = A$ weighted average of V_{front} and V_{desired} when the gap is moderate,

$V = V_{\text{desired}}$ when the gap is large

Subjects to those thresholds



Q3.3 Synchronization: Meso – Micro Boughout Interface

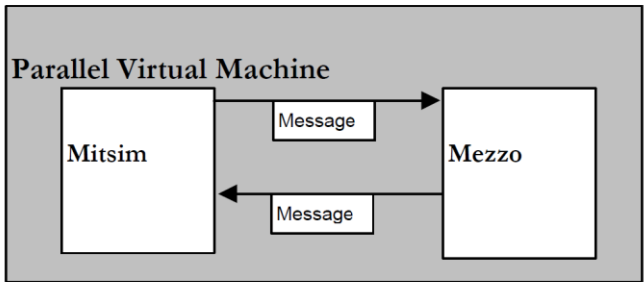


Figure 14. Message passing in the virtual machine

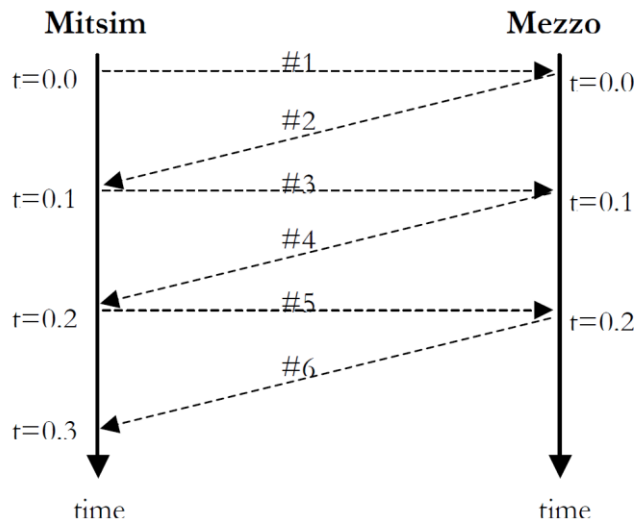
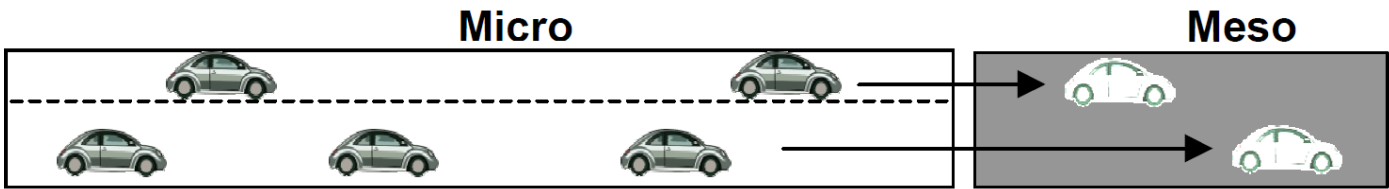
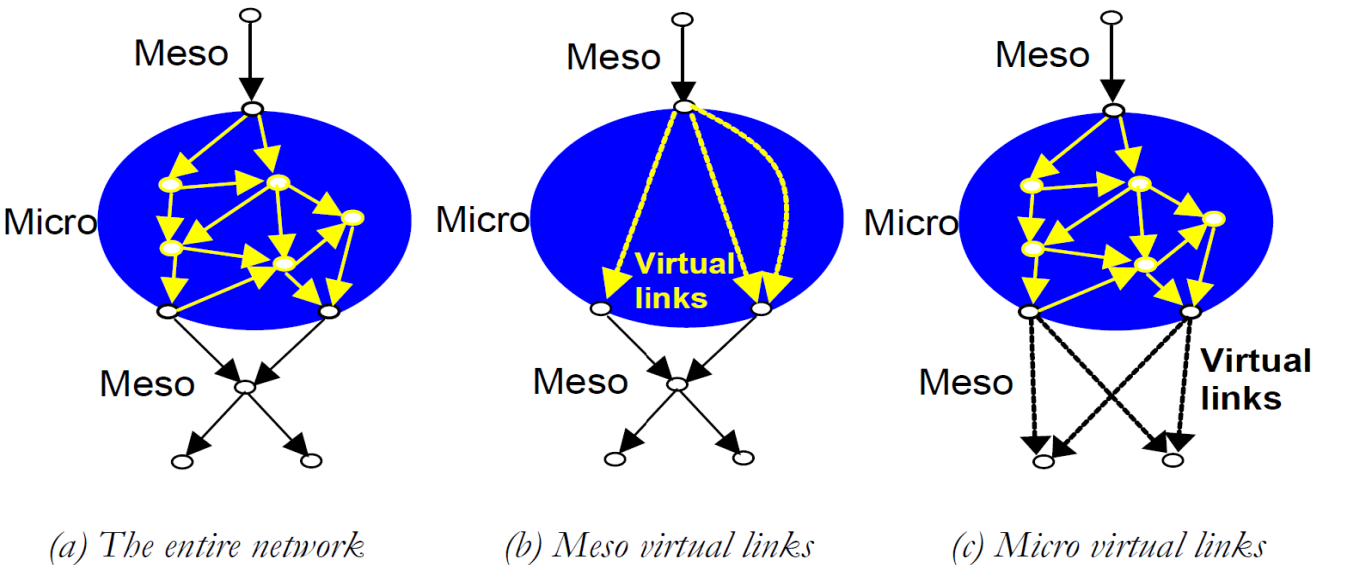


Figure 15. Event diagram of message passing between Mezzo and Mitsim



Each Micro Tick Becomes an External
Meso Event

Each micro time step enters the meso model as an external event

$$v_i(T_k + \Delta t_\mu) = \min \{v_i(T_k) + a_i \Delta t_\mu, v_{0,i}, v_{\text{safe},i}(T_k)\}$$

where

T_k = the k -th microscopic synchronization time

Δt_μ = microscopic time-step length

$a_i, v_{0,i}$ = vehicle i 's acceleration limit and desired speed

my notation; mechanism: Burghout (2004, 2005, 2006)

$$T_{k+1} = T_k + \Delta t_\mu, \quad c_k = (m_k, T_k)$$

where

T_k = the k -th microscopic synchronization time

Δt_μ = microscopic time-step length

m_k = message sent at that time

c_k = corresponding external event in Mezzo

my compact notation; mechanism: Burghout (2004, 2005, 2006)

Every micro tick produces a message, and Mezzo schedules that message as an external event at the same simulation time.

The Notion of TIME and PDES

Virtual time · Time Warp · PDES time algorithms (conservative vs optimistic)

OPTIMISTIC: break, detect, undo (Time Warp)

Rule execute speculatively on virtual time; a straggler event triggers rollback

GVT the committed floor: nothing at or below it can ever roll back

Mobility's version unexecute closures make undo cheap; asynchronous GVT

The price state saving, and rollback that can cascade

CONSERVATIVE:

Never break causality

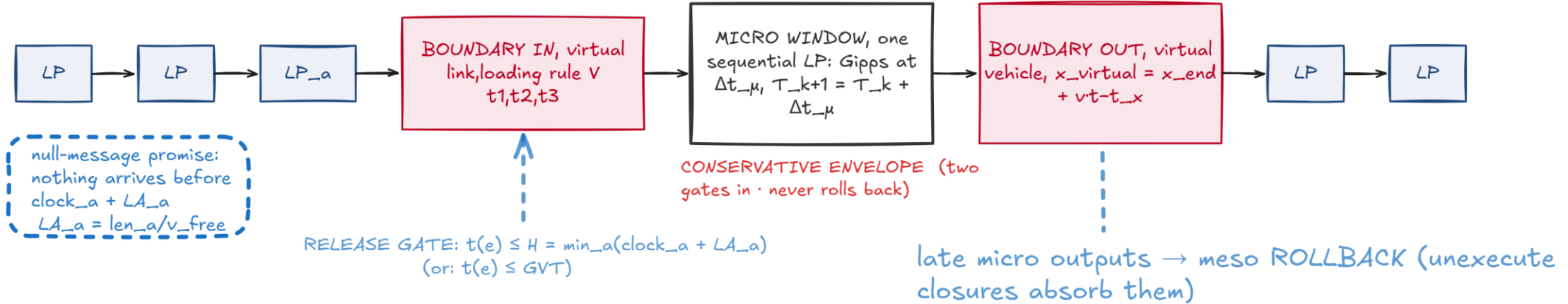
Rule Advance only when no earlier event can still arrive

The promise null messages; lookahead

Prone to lock

HPDES: Scale vs Interface definition

OPTIMISTIC MESO — Time Warp · unexecute
closures · async GVT



where:

- LP = logical process (one meso link actor)
- clock_a = current logical time of feeding link a
- LA_a = lookahead promised by link a (= len_a / v_{free})
- t(e) = timestamp of a boundary event e
- H = safe horizon: min over feeding links of (clock_a + LA_a)
- GVT = global virtual time (nothing can roll back below it)
- $\Delta t_{-\mu}$, T_k = micro time step and k-th synchronization time
- V(t₁, t₂, t₃) = multi-regime loading rule on entry headway
- x_{virtual}, t_{-x} = extrapolated leader position after exit at time

Mobiliti as the Metropolitan-Scale PDES Precedent

2018 21st International Conference on Intelligent Transportation Systems (ITSC)
Maui, Hawaii, USA, November 4-7, 2018

Links are actors (road) individual vehicles move between links as timestamped events

Scale, measured 1,110,335 nodes;
2,185,984 links; San Francisco Bay Area road network

Throughput, measured 2.38 billion events in under 30 seconds on 1,024 cores

Causal order: optimistic Algorithm, Time Warp with rollback

unexecute closures, async GVT(LP-LP talk)

Mobiliti: Scalable Transportation Simulation Using High-Performance Parallel Computing

Cy Chan, Bin Wang, John Bachan, and Jane Macfarlane
{cychan, wangbin, jdbachan}@lbl.gov, janemacfarlane@berkeley.edu
Lawrence Berkeley National Laboratory, Berkeley, CA, USA
University of California, Berkeley, CA, USA

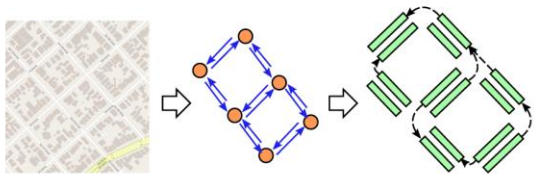


Fig. 1: Mobiliti System Model: OSM map (left) converted to road graph (nodes and links, center) converted to edge graph (right), where edges (green boxes) are the agents that send vehicle events (dashed arrows) to each other.

precede it, thus memory associated with the event and its rollback is reclaimed.

In summary, any application utilizing Devastator defines the behavior of its system agents by defining how events modify the agent's state and what (if any) future events are generated (via the *execute* function). Any modifications to the agent state must support being rolled back via the *unexecute* function, and the event supports being committed via the *commit* function.

V. TRANSPORTATION SYSTEM MODEL

A. Overall system design

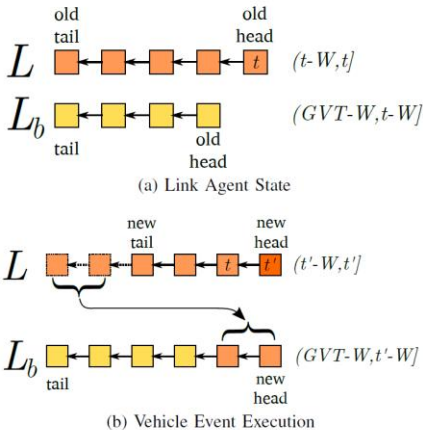


Fig. 2: Link agent state transition. Each link agent state consists of two linked lists of vehicle arrival times. When a vehicle arrival event is executed (at time t'), it is inserted onto the head of L and events older than $t' - W$ are migrated to L_b . The lists maintain the invariant that events are within the time ranges specified such that $|L|$ is the number of vehicles that arrived within time window W , and no event older than $GVT - W$ is kept in L_b .

North Los Angeles Implementation

Discuss how this could be implemented, say for North Los Angeles (the whole city may be too difficult).

North LA study area		Mobiliti,	SF Bay Area
Nodes (intersections)	28,422		1,110,335
Links (road segments)	76,307		2,185,984
Network length	11,796 km		
Study area	≈ 960 km ²		
Scale	≈ 3% of full LA	2.38B events, <30 s, 1024 cores	

measured 2026-07-10, OSMnx / OpenStreetMap drive network; Mobiliti: Chan et al.



Summary

The Agentic for Traffic Simulation with V&V

My proposal has 3 connections:

- AutoTrafficSim is the agentic experimentation layer with explicit:
 - V&V gates, structured feedback, lineage, and a human decision boundary.
 - The scientific application is I-210 and SR-134 wildfire evacuation and contraflow, tested across visible behavioral and hazard assumptions.
 - The simulation foundation is on Scalation
 - Wildfire Model building and unit testing
- Motivated to test the HPDES scenario as a separate finding
- With my Major advisor Approval (Dr John Miller) Formalize V&V Concept for Discrete Event Traffic Simulation Community in the age of Artificial Intelligence

Thank you for listening to this talk

Question and Answer



Daganzo Cell Transmission Model

$$n_i(t + 1) = n_i(t) + y_i(t) - y_{i+1}(t)$$

$$y_i(t) = \min\{ n_{i-1}(t), Q_i(t), N_i(t) - n_i(t) \}$$

where

$n_i(t)$ = number of vehicles in cell i at time t

$y_i(t)$ = inflow into cell i during the interval after t

$Q_i(t)$ = inflow capacity of cell i

$N_i(t)$ = holding capacity of cell i

Daganzo (1994), Cell Transmission Model